EPSR-test-ex86 dockato: Qv bonding sites

Program(s): dockato | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

`dockato` [1] builds larger molecules from `.ato` building blocks: copies of the first molecule of a docking file are bonded onto free sites of a docked file, removing one atom of each to form the new bond. EPSR-style dummy atoms (`Qv`) can mark the bonding sites; they are removed on docking and their atom type disappears.

Code exercised

Fortran (original, fortran_src/):

- `dockato.f`: `dockato` (driver, parameters from command line or stdin), `rotatem`, `random_rotatem`, `search_unsaturated`, `find_valence`
- `dockato.f`: `test_overlap`, `bond_molecule` (ROT/DIH groups), `replace_references`, `add_neighbours`, `remove_atoms`, `clean_atom_types`
- `ato_manipulation_rev4.f`: `get_direct_bonds`, `get_rotgrp`, `centreofmass`, `ftype`; `ato_read_and_save.f`: `moloutc`

Rust (port, rust/epsr-core/src/):

- `tools/dockato.rs`: `run`, `rotatem`, `random_rotatem`, `search_unsaturated`, `find_valence`, `test_overlap`
- `tools/dockato.rs`: `bond_molecule`, `replace_references`, `add_neighbours`, `remove_atoms`, `clean_atom_types`, `get_direct_bonds`, `get_rotgrp`
- `ato.rs`: `State::moloutc` (ROT/DIH groups), `State::ftype`; `rng.rs`: `Ran1::ran1`

Physical system

A water molecule (O-H 0.976 Å, SPC/E parameters [2]) with an extra invisible dummy atom `Qv` bonded to O1 at 1.45 Å. Atom `atomd` of the new molecule is placed on `atomv`, the molecule is rotated so that its `atomdr` neighbour lies on the `atomvr` neighbour of `atomv` (equal bond lengths required), an optional second (dihedral) rotation about the new bond uses `atomdrr`/`atomvrr` (random angle in [dminrr, dmaxrr] from `ran1`, the Park-Miller generator with Bays-Durham shuffle [3,4], or a fixed dihedral that creates a DIH group when dminrr = dmaxrr), overlaps are tested with a distance factor, `atomd` and `atomvr` are removed and the new rotational group, bonds and atom types are updated. Here Qv of the docking molecule goes onto the docked O1 and the docking O1 onto the docked Qv; both Qv atoms are removed and an O-O bond of 1.45 Å joins the two units.

Inputs

`WaterQv.ato` created from `WaterMolecule.ato` (Qv atom and Qv atom type added). Run 1: `Qv O1 H1 2 10 WaterQv.ato O1 Qv H1 0 360 1.0 peroxide`; run 2: one docking with fixed dihedral 110° (DIH), `nooverlaps`, answers from `fixed.stdin`.

How the accuracy was verified

`commands.txt` (2 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Each run writes one new `.ato` file (coordinates, bonds, rotational groups, atom types and the random-number state), compared byte for byte with the stored Fortran output. Both outputs hold 6 atoms (two O1, four H1) in an 18.17 Å box; `peroxide_dih.ato` has a DIH group with the dihedral 110.000. In run 1 the second docking finds no free site (restore path).

Notes

Fortran behaviour reproduced: the failed second docking's `ftype` call re-creates the Qv type, which stays in `peroxide.ato` after `clean_atom_types` removed it.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).

[3] S. K. Park and K. W. Miller, "Random number generators: good ones are hard to find", Commun. ACM 31, 1192-1201 (1988).

[4] W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, Numerical Recipes in Fortran 77, 2nd ed., Cambridge University Press (1992).

[5] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.